

Thank you for purchasing a FANUC Robot

READ THIS FIRST

Use this guide to unpack, install, turn on, and jog your new FANUC robot.

The appropriate level of safety for your application and installation can best be determined by safety system professionals. A qualified electrician should apply power. FANUC recommends that each customer consult with such professionals in order to provide a safe application, and take the necessary FANUC training course to operate the robot safely. Refer to your FANUC documentation for safety procedures.

Required Tools

- ◆ Cross-head screwdriver ◆ Box cutter
- ◆ Metric hex-head wrenches

Required Items

- ◆ Customer-supplied properly rated power source as specified on the Controller Power Source Label

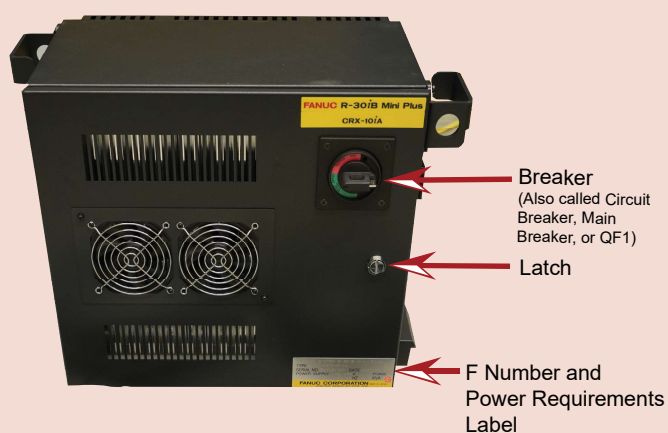
- ◆ Rigid platform to hold robot

Optional Items

- ◆ Robot locator pins ◆ USB with robot software *

* **Note:** To download software to a USB, go to your regional FANUC customer service software downloads website.

Controller



Tablet Teach Pendant



Robot (mechanical unit)

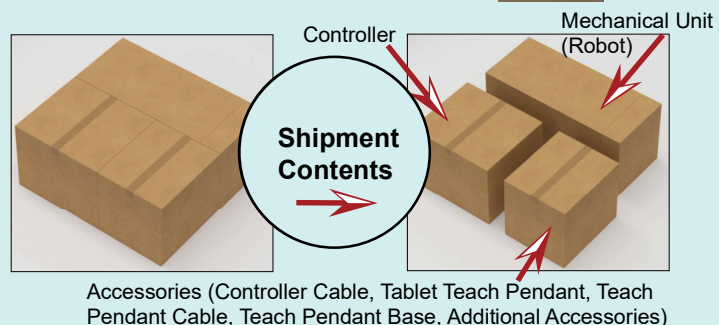
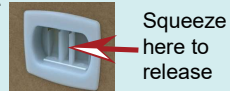


CRX-10iA variant shown

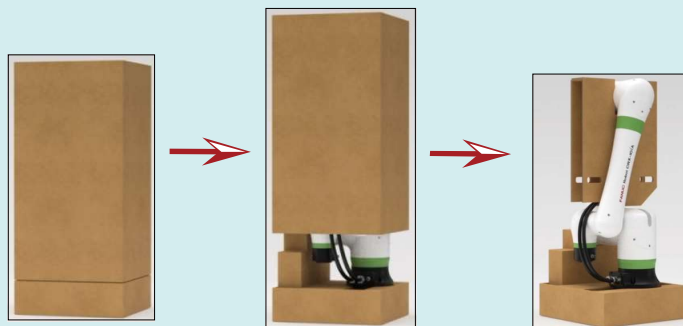
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A. Unpack/Install your Robot

1. Remove the cardboard cover by squeezing the packing clips and lifting the cover up. Underneath the cover you will find three boxes.



2. Locate and open the box containing the robot and connection cable.



3. Using the contact lift points, lift the robot out of the box and place it in its new location.

CAUTION

Be careful lifting the robot. The robot weighs approximately 40 kg. FANUC recommends using two people to lift the robot safely.



WARNING

Once the robot is in place, continue to support the robot until it is fastened to its base plate or platform.

4. Fasten the robot to the base plate or platform using the (4) M8 bolts and washers supplied.



Robot installation is complete. You are now ready to go to **B. Unpack/Install your Controller**

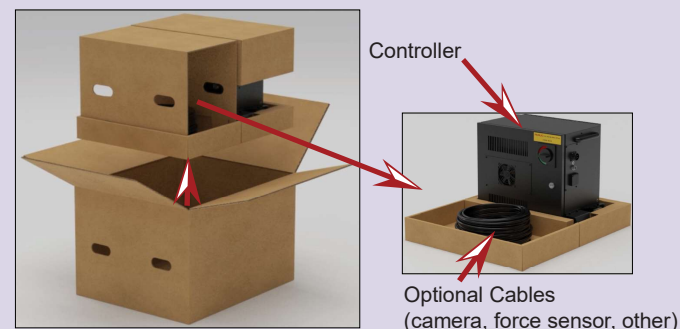
B. Unpack/Install your Controller

1. Remove the controller and cables from the box.

Note: The controller has handles on the sides to help with lifting.

CAUTION

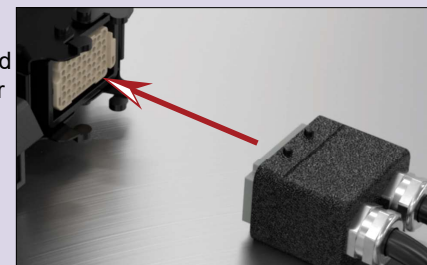
Be careful lifting the controller. The controller weighs approximately 20 kg.



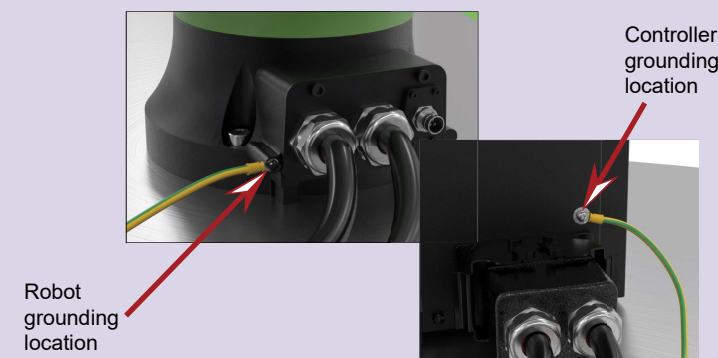
2. Place the controller in a location such that the grounding wire and robot connection cables can reach from the robot to the controller. Leave enough space to open the controller door.



3. Plug the robot cable into the controller and secure the connector with the clips.

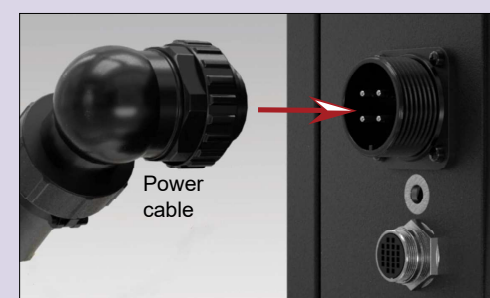


4. Retrieve the ground cable from the accessories box. Connect one end to the robot and the other end to the controller.

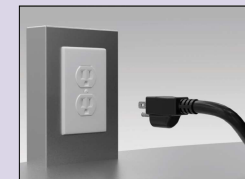


Note: The larger ring terminal attaches to the controller.

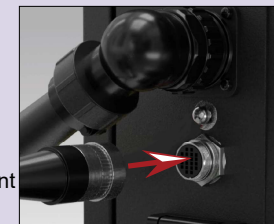
5. Connect the robot power cable to the controller.



6. Plug the power cable into a properly rated power source (AC 100-120V, Single-phase, 50/60Hz)



7. Connect the teach pendant cable to the controller as shown to the right.



8. Connect the teach pendant cable to the teach pendant as shown to the right.

Teach pendant cable



Controller installation is complete. You are now ready to go to **C. Turn on your Controller and Prepare to Jog**

C. Turn on your Controller and Prepare to Jog

WARNING

Lethal voltage is present in the controller whenever it is connected to a power source. Be extremely careful to avoid electrical shock. Lock out and tag out the power source at the controller according to the policies of your plant.

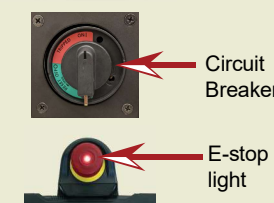
1. Power on the tablet by pressing and holding the top button on the right side of the tablet. When the tablet has powered up, you should see "connecting".



ON/OFF Button

2. Turn on the controller by rotating the Breaker clockwise to ON.

Note: The e-stop light on the teach pendant base unit will light when the Breaker is ON and power is connected to the controller.

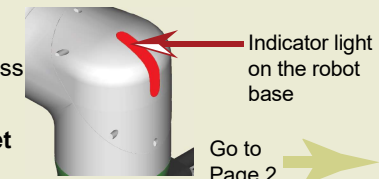


E-stop light

3. Once the indicator light on the robot base turns ON, the tablet should automatically begin communicating with the robot. If not, touch the **Tablet TP** icon on the tablet to open the CRX interface.

Note: If the **Tablet TP** icon is not available:

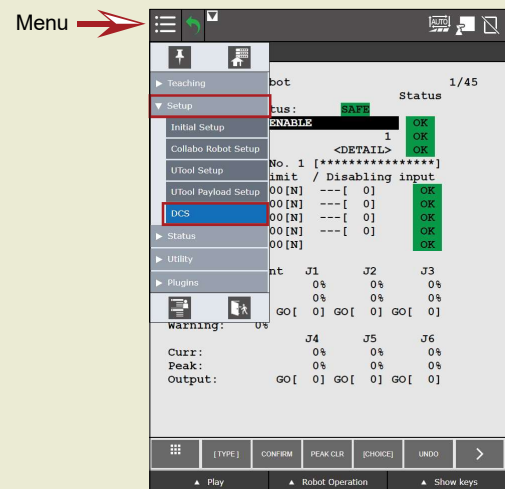
- a. Swipe up on the Home screen to access the All Apps screen.
- b. Press and hold **Tablet TP** to add it to the Home screen.



Go to Page 2

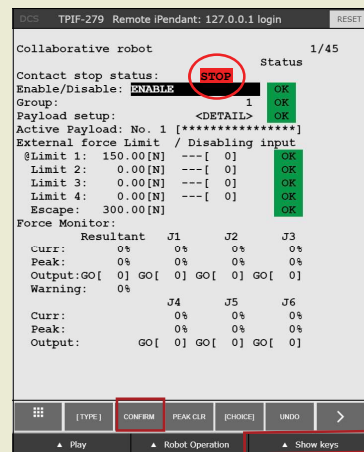
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4. If the DCS Collaborative robot screen does not display after power up, touch the Menu icon  and select **Setup > DCS**.



5. Press **Show keys**.

6. Press **CONFIRM** and input the Code number, "1111".



7. When prompted, confirm that the **Active Payload** number is 1 for the 0 kg schedule and press **YES**.

Note: If the tool is attached to the robot, ensure accurate payload data is entered and the appropriate payload schedule is confirmed.

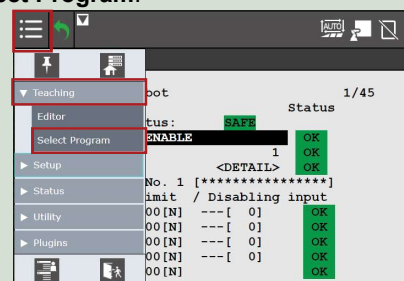
8. When prompted, confirm "Nobody contacts with the robot?" by pressing **YES**. You will see "Payload confirmation success."

Note: Contact stop status will change from STOP to SAFE.

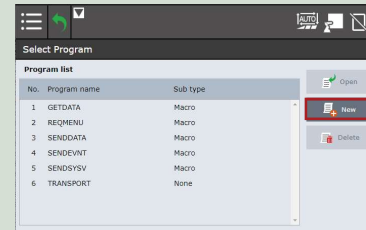
You have successfully turned on your controller.
You are now ready to go to **D. Jog your Robot**.

D. Jog your Robot

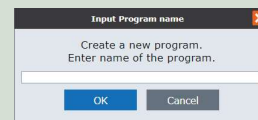
1. From the Menu icon , select **Teaching > Select Program**.



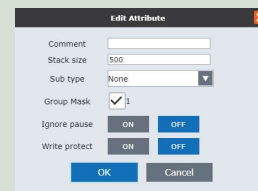
2. Create a new teach pendant program by pressing **New**.



3. Enter the program name and press **OK**.



4. Edit attributes, if desired and press **OK**.



5. Enable the tablet teach pendant by toggling the icon located at the top right of the screen.

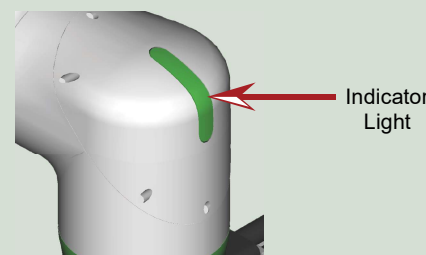


6. Ensure the Emergency Stop button on the teach pendant is popped out. If not, twist the button clockwise to reset it.

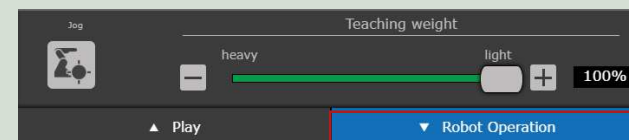
7. Reset any faults by pressing the **RESET** button located at the top-right of the Editor.



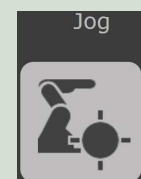
Note: The indicator light on the robot will turn green and the servo motors will turn ON when the faults are cleared.



8. Press **Robot Operation**.



Note: On the left of the screen, one of two icons will be visible. The visible icon indicates the screen you want to view, not the screen you are currently viewing.



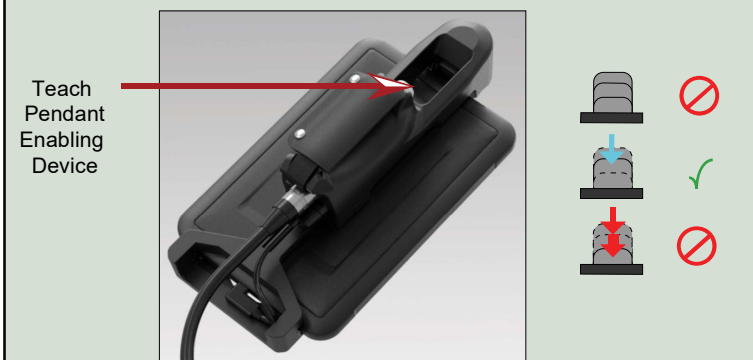
Press if you want to go to the jog screen



Press if you want to manually guide the robot (by hand)

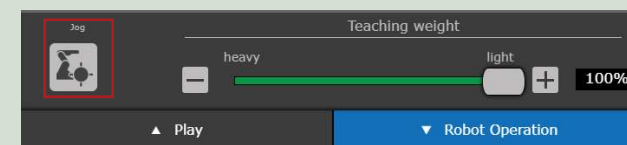
9. To move by hand, make sure the **Jog** icon is visible to manually guide the robot.

10. Press the Enabling Device to the middle position.

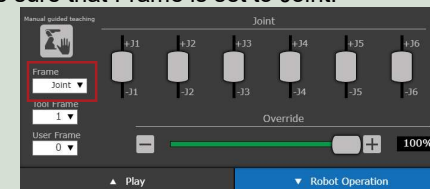


11. While pressing the Enabling Device, grab the robot and move it to a given position manually (by hand).

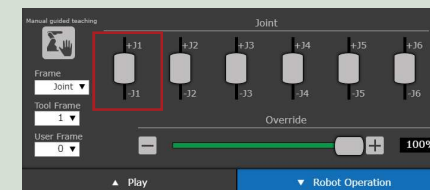
12. To jog each joint individually using the teach pendant, first press the **Jog** icon to activate the screen for joint jogging.



13. Make sure that **Frame** is set to **Joint**.



14. Use the directional slider to move each joint individually. For example, to move J1 in the positive direction press the J1 slider up. To move J1 in the negative direction press the J1 slider down.



15. To jog using Cartesian coordinates (XYZ), change the **Frame** from **Joint** to **Cartesian**.



16. Slide the Position buttons to move X, Y, or Z, and the Orientation buttons to "roll around" X, Y, and Z.



Jogging the robot is complete.
If you need to update your software, go to **E. Software Loading**

E. Software loading Page 2

- Place the USB with Robot Software into UD1 located on the side of the robot controller.
- Press  > **Setup > Initial Setup** to access the **Initial Setup** guide on the Tablet TP.
- Proceed through each step of the **Initial Setup** guide.
- When prompted to begin the software update in the **Initial Setup** guide, select **Update now**.

- First the Boot Monitor (Controller firmware) will update. This takes a minute or two. **Do not power off the controller.**

Note: If **File Not Found** or an **HRTL-060 Ethernet connection timed out** alarm is displayed, close and reopen the Tablet TP App by pressing the Back (<) button. This is normal; the Boot Monitor still updated successfully.

- When the Tablet TP App is reconnected, make sure that it displays a message to cycle the power on the controller.

Note: You might need to complete the final step of the **Initial Setup** guide before you are prompted to cycle power.

When prompted to cycle power, power off the controller, wait 10 seconds and power on again to start the software installation. The Tablet TP App will lose connection to the controller. The entire software installation process could take 1 to 2 hours. **Do not power off the controller while software installation is in progress.**

- When the software installation is complete, the light on the J1 base will turn on. Please wait for a few seconds for the Tablet TP App to connect to the robot controller.

Note: If the Tablet TP App does not reconnect a few seconds after the light on the J1 base turns on, close and reopen the application. You might also need to clear cache on the Tablet TP App before it can reconnect.

- If the **TPIF-270 Clear Browser Cache** alarm occurs, it indicates that the app's cache needs to be cleared before reconnecting. To clear cache:

- Close the app and return to the tablet Home screen
- Press  (Settings) -> **Apps** -> **Tablet TP**
- Select **Storage**
- Press **Clear cache**

- If you want to add the Tablet TP App to the Home screen, swipe up on the tablet's Home screen to access the ALL Apps screen. Press and hold **Tablet TP** to add it to the Home screen.

Congratulations! Installation of your CRX robot is complete.

For more information

To access controller, robot, or application-specific manuals, scan the **QR Code** to the right or go to <https://www.fanucamerica.com/qr/edocs>.



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Canada: 1-905-812-2300

Mexico: + 52 (449) 922-8000

South America: + 55 (11) 3619-0599

Or, select your region at <https://www.fanuc.com>.

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